QBRAID CHALLENGE · QISKIT FALL FEST CT 2026

Instant Insanity on a quantum computer

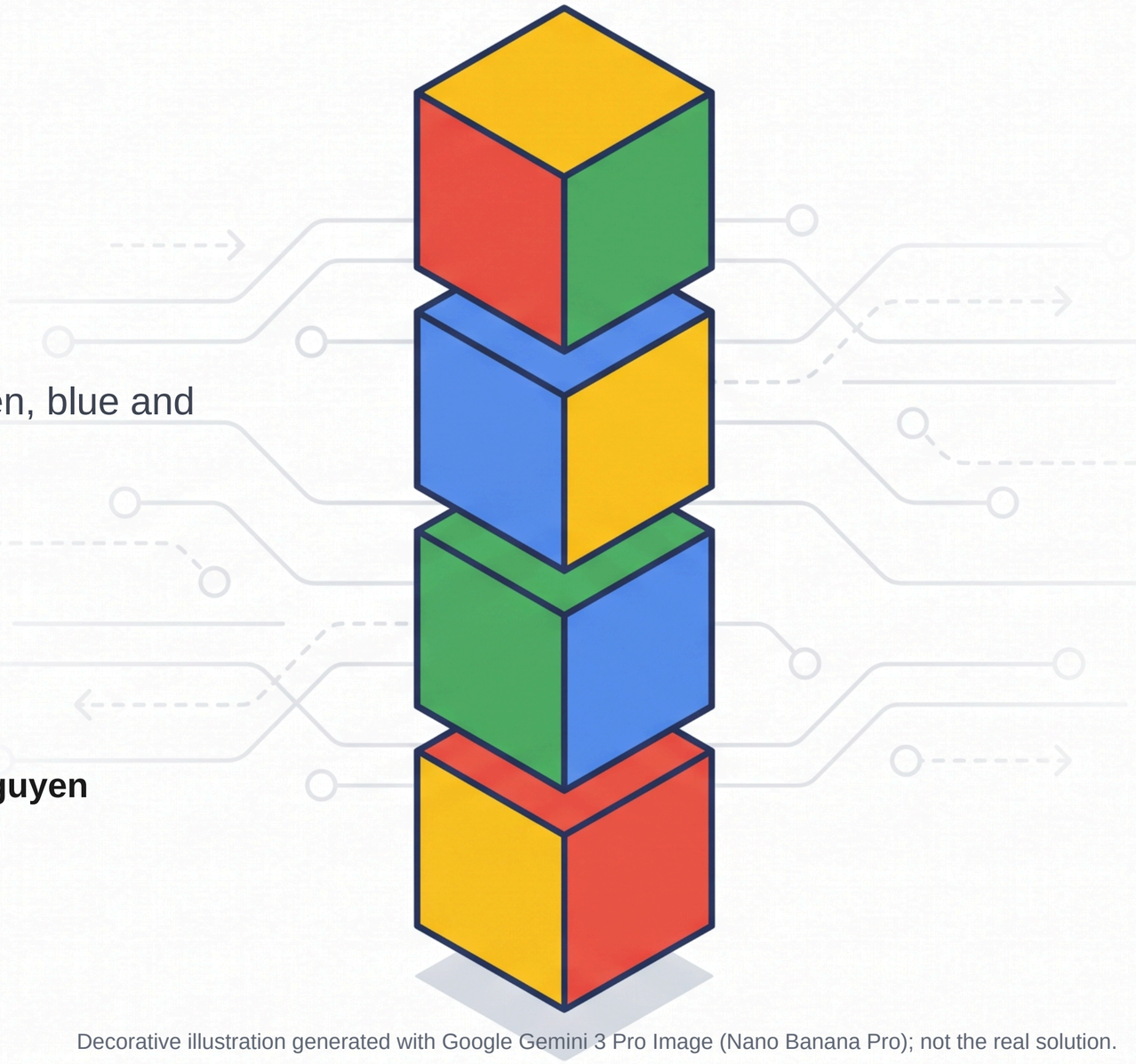
Stack 4 cubes so every side of the tower shows red, green, blue and yellow exactly once.

331,776
ways to stack the cubes

8
of them work
(one tower, 8 views)

12
qubits in our search

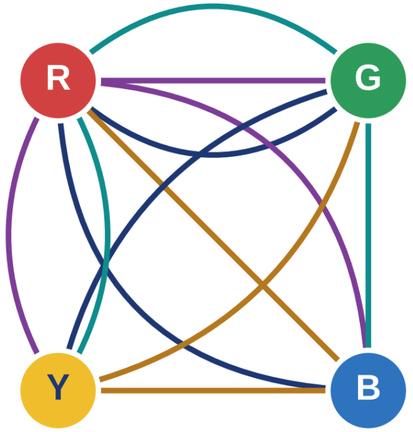
Kiarash Naghavi Khanghah, Luis A. Mendez, Hoang Anh Nguyen
Core and bonus levels (N cubes, N colours)



Decorative illustration generated with Google Gemini 3 Pro Image (Nano Banana Pro); not the real solution.

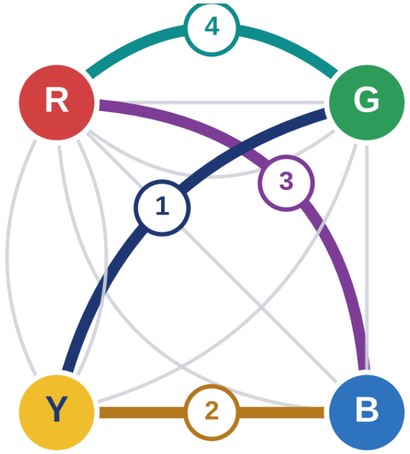
THE PUZZLE AS A GRAPH

All 12 opposite-face pairs



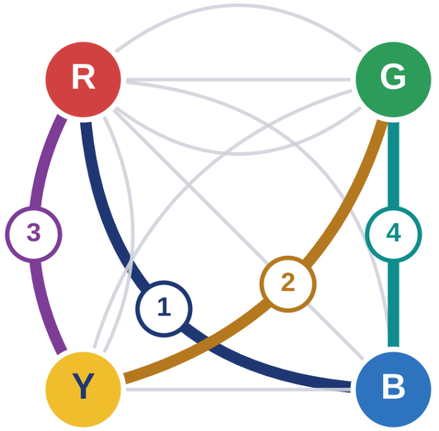
one line per pair, coloured by cube

Front / back pick



1 line per cube, every colour touched twice

Left / right pick



4 other lines, same rule

cube 1 cube 2 cube 3 cube 4

Solution = two picks with no shared line, each giving every colour exactly 2 lines

1 line = 1 pair of opposite faces.
Each cube has **6 possible placements**.

6^4 = 1,296 cases
instead of 331,776 stackings.

2 cases solve it:
the tower and its quarter turn.

ENCODING: 3 QUBITS PER CUBE

A cube's placement = a number 0 to 5

cube 1 has pairs A = GR, B = BR, C = GY

bits	code	front/back	left/right	hidden
000	0	A G R	B B R	C
001	1	A G R	C G Y	B
010	2	B B R	A G R	C
011	3	B B R	C G Y	A
100	4	C G Y	A G R	B
101	5	C G Y	B B R	A
110	6	not a placement, never produced		
111	7	not a placement, never produced		

12 qubits

for 4 cubes. The starter's one-hot encoding needs 24.

0.15 %

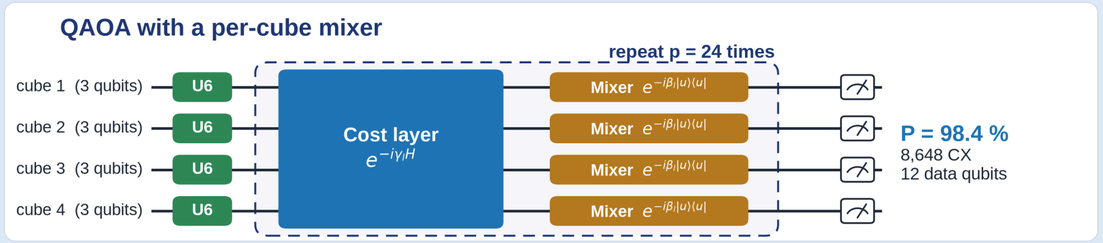
chance that a random guess is the tower. This is the bar to beat.

Checked first

The rules' cost is zero on exactly 2 of the 4,096 bit strings: the two answers.

TWO QUANTUM SEARCHES, SIDE BY SIDE

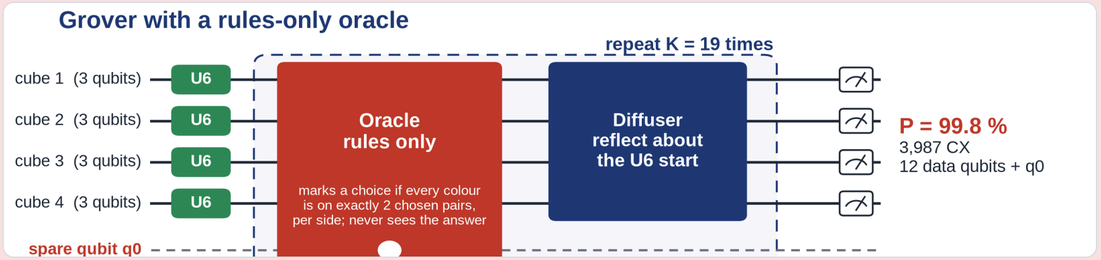
QAOA



A **round** = score each combination by the rules it breaks, then shuffle each cube among its 6 placements.

Generalizes: the same angles work for 5 to 8 cubes.

Grover

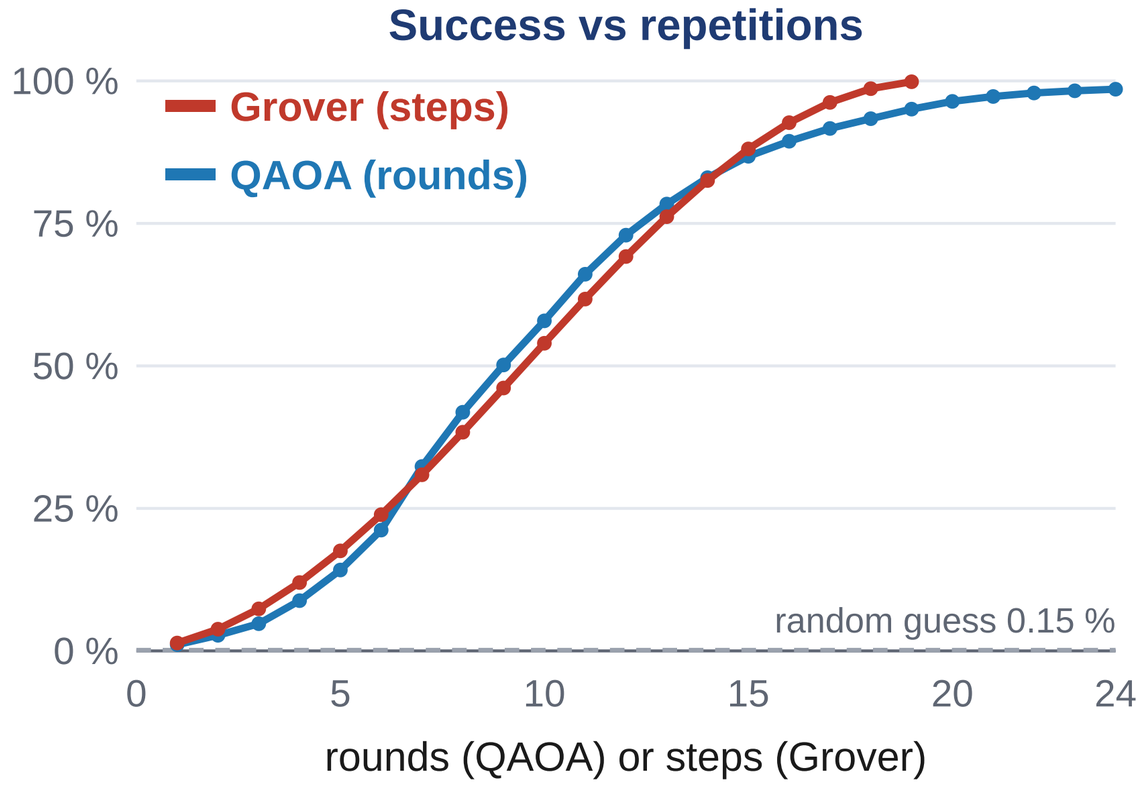


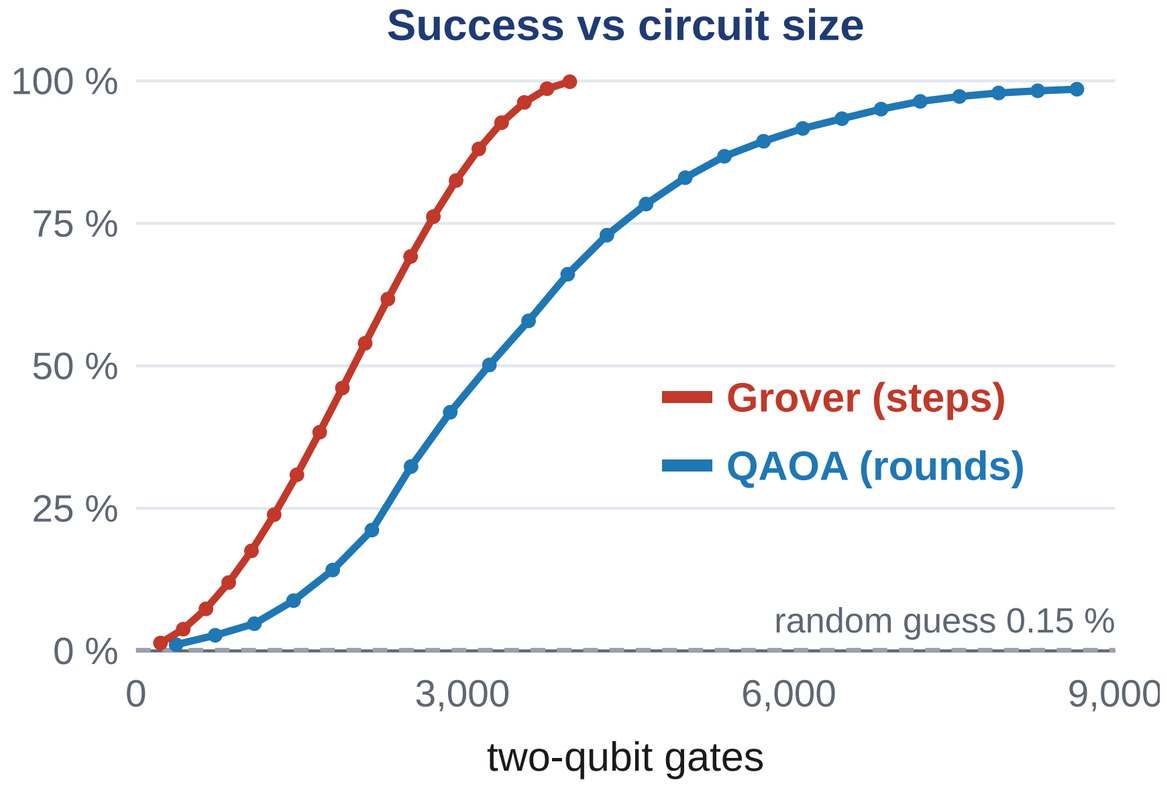
A **step** = a checker built only from the rules marks the solutions, then an amplifier boosts them.

Wins for 4 cubes: fewer gates, higher success.

More rounds or steps = a deeper circuit = higher success on a perfect machine.

MORE ROUNDS OR STEPS, MORE SUCCESS





Grover reaches 99.85 % in 19 steps; QAOA reaches 98.5 % in 24 rounds. Random guess: 0.15 %.

4-cube tower, exact noiseless probabilities; two-qubit gates counted after transpiling.

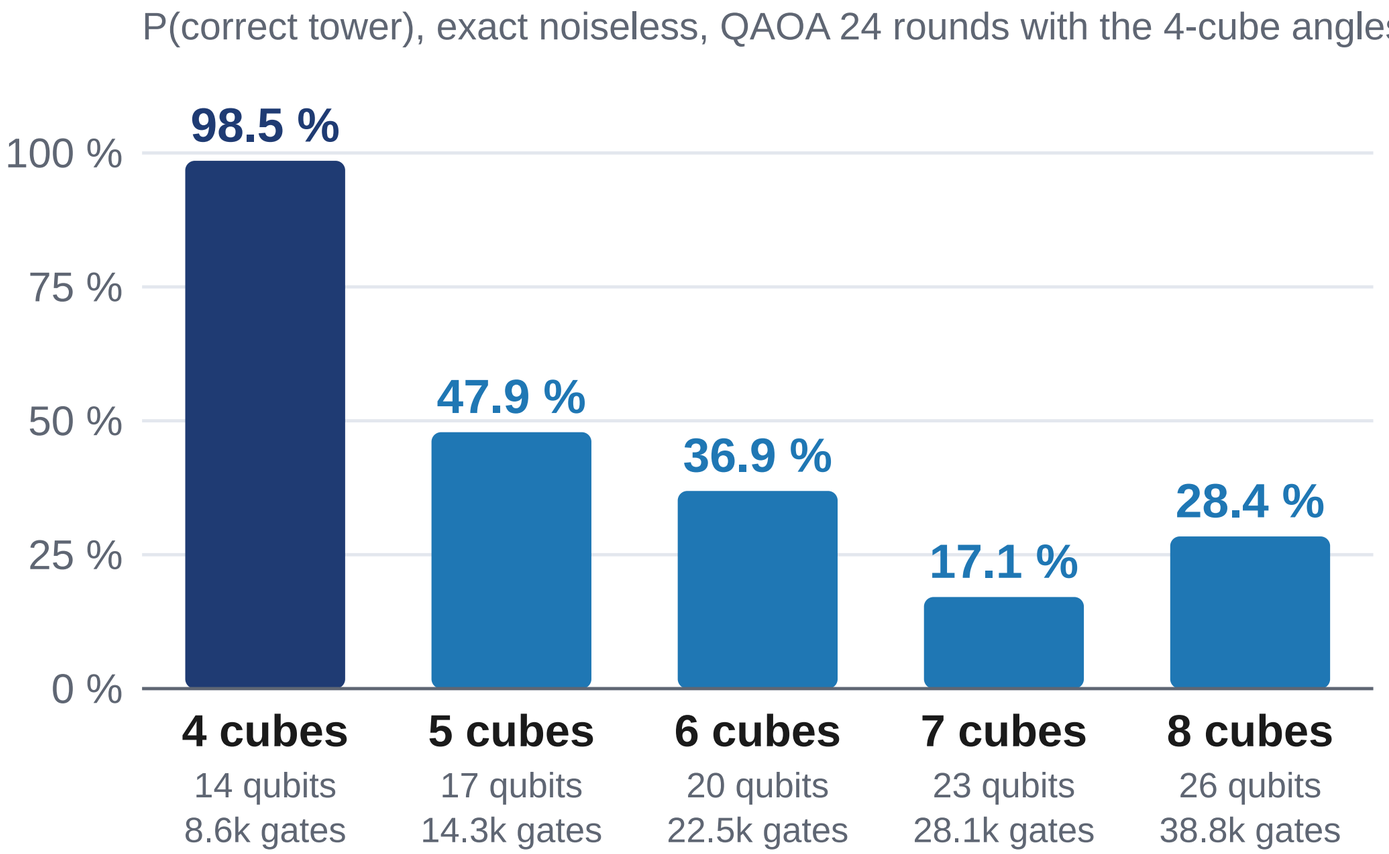
RESULTS: SIMULATOR, CLOUD AND REAL HARDWARE

Experiment	two-qubit gates	Perfect simulator	qBraid SV1	Rigetti QPU	Random guess
4-cube tower: Grover					
4-cube tower, Grover 5 steps	1,061	17.5 %	–	–	0.15 %
4-cube tower, Grover 10 steps	2,106	54.0 %	–	–	0.15 %
4-cube tower, Grover 19 steps	**3,987**	**99.85 %**	**99.95 %** ± 0.04	–	**0.15 %**
4-cube tower: QAOA					
4-cube tower, QAOA 3 rounds	1,088	4.8 %	–	–	0.15 %
4-cube tower, QAOA 12 rounds	4,328	72.9 %	–	–	0.15 %
4-cube tower, QAOA 24 rounds	**8,648**	**98.5 %**	**98.6 %** ± 0.2	–	**0.15 %**
4-cube tower, QAOA 24 rounds, angles trained on another puzzle	8,648	52.1 %	–	–	0.15 %
4-cube tower, starter 24-qubit encoding, QAOA 3 rounds	940	5.2 %	–	–	0.00001 %
Bonus					
Bonus 5-cube tower, QAOA 24 rounds	**14,266**	**47.9 %**	**46.6 %** ± 1.1	–	**0.026 %**
Bonus 6-cube tower, QAOA 24 rounds	22,476	36.9 %	–	–	0.0043 %
Bonus 7-cube tower, QAOA 24 rounds	28,142	17.1 %	–	–	0.00071 %
Bonus 8-cube tower, QAOA 24 rounds	38,752	28.4 %	–	–	0.00012 %
Real hardware (Rigetti QPU)					
4-cube tower, Grover 1 step	225	1.4 %	–	**0 / 500**	0.15 %
Bonus 5-cube tower, Grover 1 step	287	0.23 %	–	**0 / 500**	0.026 %

Works on a perfect simulator and on qBraid SV1; today's QPU loses the signal even at 225 two-qubit gates.

Perfect simulator = exact noiseless value. Deeper circuits (QAOA 24 rounds, 8,648 gates) were rejected by the QPU. ± = one standard error. Every success re-checked by `solves()`. Job IDs in `results/jobs.json`.

BONUS: 4 TO 8 CUBES WITH THE SAME ANGLES



Still works: 17 to 48 % success at 5 to 8 cubes, with no retraining.

Grows: 3 qubits and 5,600 to 10,600 two-qubit gates per extra cube.

Honest: brute force is still faster. No speedup claimed.

Next: quantum counting, so Grover finds the number of answers itself.